Weekly Update

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Last Update: October 19, 2026

1 Main Findings

1.1 DB-Sorter

quick background. DB-sorter uses the sorting characteristic of DB flow to find ground state or excited states of a hamiltonian H using a diagonal matrix D that has an engineered order in its diagonal elements (usually increasing).

Findings:

- I chatted with ChatGPT and it annihilated my work so far. Apparently what I have needs more work. Here is a list of suggestions:
  - Only proposing a few "good" D's is not that good. Let's see more theoretical work on how a (near-)optimum D look like
  - There is not enough work done regarding excited state, despite the claim is it can also address them
  - More work on sorting convergence can be done besides simply fixed point
  - Also compare against methods like subspace-VQE and folded spectrum
- Optimum D can be roughly found using Lagrange multipliers and a variational circuit. However my method so far is exponentially expensive. See 3.1.1

2 Plan For Next Week(s)

2.1 DB-sorter

- Finding a more efficient way of finding optimum D.
- More analysis on excited states and sorting behaviour.
- Comparison with subspace-VQE and folded spectrum.

3 Details

3.1 DB-sorter

3.1.1 Best D

Optimally, we want the path from $H(0)$ to $H(\infty)$ be the shortest, which means minimizing

$$\int_0^\infty \|dH\|$$

However, on the quantum circuit, we only implement the first few steps. Therefore in a greedy way, we want to maximize the change from H_0 to H_1 or in other words:

$$\begin{aligned}
\max\|dH\| &= \max_D tr([[D,H],H]^2) && (1)\\
&= \max_D tr(6H^2DH^2D - 8H^3DHD + 2H^4D^2) && (2)\\
&= \max_{d_1,\ldots,d_n} \sum_{k=1}^{n} d_k^2(6H^2[k,k]H^2[k,k] - 8H^3[k,k]H[k,k] + 2H^4[k,k]) && (3)\\
&= \max_{d_1,\ldots,d_n} \sum_{k=1}^{n} d_k^2 m_k && (4)
\end{aligned}$$

Our implementation of bracket relies on $e^{-s[D,H]} = e^{\sqrt{s}D}e^{\sqrt{s}H}e^{-\sqrt{s}D}e^{-\sqrt{s}H}$. In this implementation we have two observations: a) adding a constant value to all elements of D would not affect the final results and b) multiplying a constant to all elements of D can be equivalent of changing our step size. These two observations let us assume that all elements of D are in $[0,1]$. We try two scenarios:

Case I: $d_1 = 0 \leq d_2 \leq \cdots \leq d_n = 1$ and Case II: $d_1 = 0 \leq d_2, d_3, \ldots, d_{n-1} \leq d_n = 1$. Both of these can be solved using KKT Lagrange multipliers.

For case I, the problem description would be:

$$\begin{aligned}
&\max \sum_{k=1}^{n} d_k^2 m_k && (5)\\
&\text{s.t } d_k \geq d_{k-1} : k \geq 2 && (6)\\
&d_1 = 0, d_n = 1 && (7)
\end{aligned}$$

which results in the Lagrangian:

$$L(\{d_k\}_1^n, \{\lambda_k\}_2^n) = \sum_{k=1}^{n} d_k^2 m_k - \sum_{k=2}^{n} \lambda_k(d_{k-1} - d_k) \quad (8)$$

solution:

$$\begin{aligned}
&2d_k m_k - \lambda_{k+1} + \lambda_k = 0 \Rightarrow d_k = \frac{\lambda_{k+1} - \lambda_k}{2m_k} && (9)\\
&\lambda_k(d_k - d_{k-1}) = 0 && (10)\\
&\lambda_k \geq 0 && (11)
\end{aligned}$$

Considering the complementary slackness, we can test different number of starting k's for which $\lambda_k = 0$. If the first r Lagrange multipliers are zero, from above equations we will have

$$d_1, \ldots, d_{r-2} = 0, d_{r-1} = \frac{\lambda_r}{2m_r}$$

With induction and complementary slackness, you can show that (except for an edge case), all feasible solutions have $\lambda_{k>r} = \alpha\lambda_r, \alpha > 0$, therefore $d_{k\geq r-1} = 1$. This means that the solution to this optimization is a diagonal matrix with the first r elements equal to 0 and the rest equal to 1.

For case II, the problem description would be:

$$\max \sum_{k=1}^{n} d_k^2 m_k \tag{12}$$

$$\text{s.t } 0 \leq d_k \leq 1 \tag{13}$$

$$d_1 = 0, d_n = 1 \tag{14}$$

which results in the Lagrangian:

$$L(\{d_k\}_1^n, \{\lambda_k\}_1^n, \{\gamma_k\}_1^n) = \sum_{k=1}^{n} d_k^2 m_k - \sum_{k=1}^{n} \lambda_k (d_k - 1) + \sum_{k=1}^{n} \gamma_k d_k \tag{15}$$

solution:

$$2d_k m_k - \lambda_k + \gamma_k = 0 \Rightarrow d_k = \frac{\lambda_k - \gamma_k}{2m_k} \tag{16}$$

$$\lambda_k (d_k - 1) = 0 \tag{17}$$

$$\gamma_k d_k = 0 \tag{18}$$

$$\lambda_k, \gamma_k \geq 0 \tag{19}$$

Using complementary slackness, we should have all elements arbitrarily 0 or 1.

Both cases boil down to exponentially many candidates to check the maximum in. Also computing the trace in equation 2 requires exponentially many Pauli terms.